

Funded Ratio Risk in Public Pension Plans

Group 21

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Question of Interest

Our goal is to model funding shortfall risk for local and state governments employee pension plans.

This is an important question because tens of millions of Americans rely or will rely on these government pension plans for their retirements.

These pension funds are also a significant asset class in the United States financial system, with over 6 trillion dollars in assets for local and state governments.

A strong individual example comes from the New York City Police Department Pension Fund, which alone manages over 60 billion dollars in assets.

Public Plans Data

Our data comes from the Public Plans Dataset from Boston College.

It includes quarterly time series data for almost 250 major state and local government pension plans.

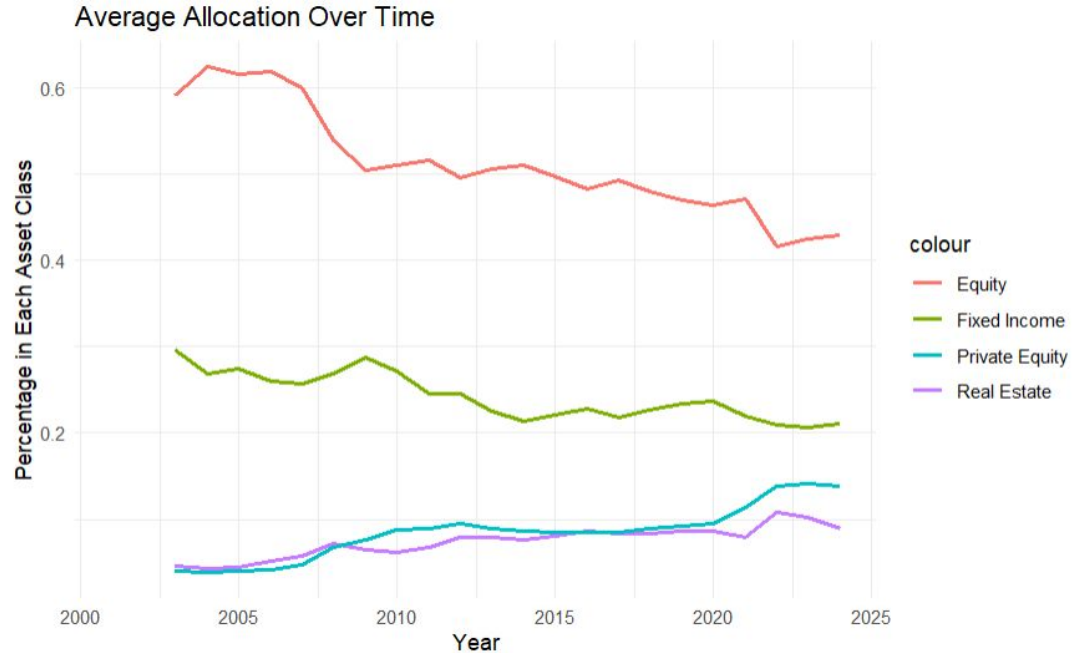
The dataset covers 95% of both pension plan membership and assets and is split roughly 50/50 between state and local pension plans.

Asset and liability holdings only reported annually, so we fit models to class representative ETFs to obtain monthly return data.

Asset Allocation

The graph on the right represents a weighted average of the asset allocations for the entire dataset.

Over time, pension funds moved into real estate and private equity while decreasing their holdings in equity and fixed income securities.

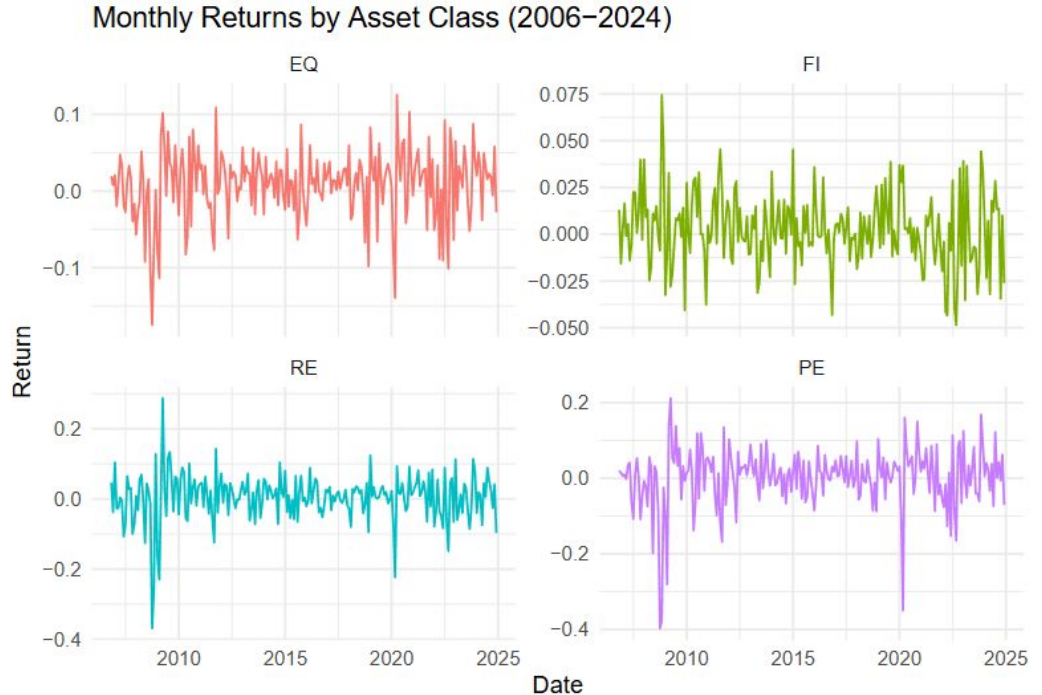


Asset Returns

On the right is a graph of monthly returns grouped by asset class for the pension funds.

There is clear instability around the 2008 and 2020 recessions.

Private equity has the highest average return over the period, followed by equity, real estate, with fixed income securities being the lowest.



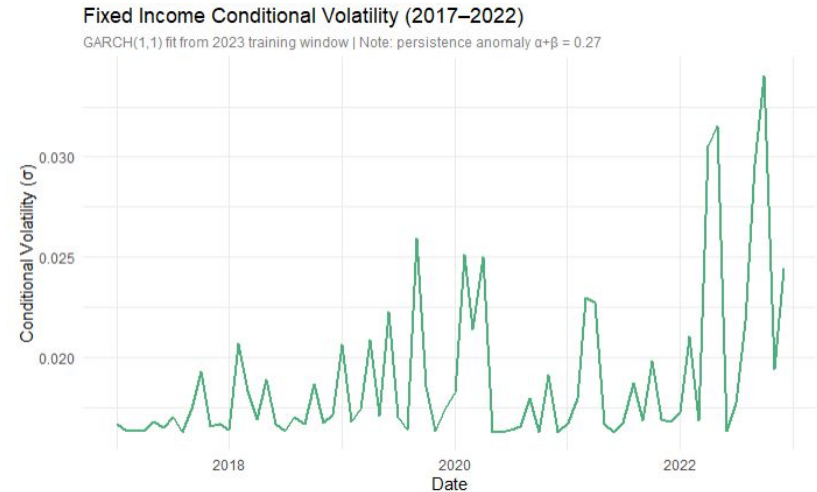
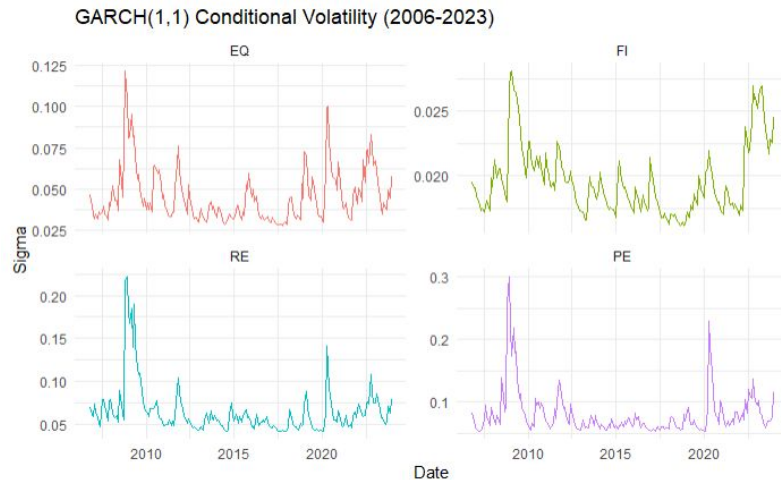
Prediction Methodology

- model asset class returns using a GARCH(1,1) with skewed t-student errors to capture volatility.
- fit standardised residuals to a t-copula, with an expanding window from 2006 to each test year, to account for correlation.
- Simulate 10,000 Monte Carlo paths propagating the correlation\volatility :
 - annual rebalancing
 - add net flows
 - value implied yield changes from IEF returns
 - use constant industry standard duration ~13
- Compare VaR/cVaR to constant volatility assumptions.

GARCH Modelling and Diagnostics

All four assets experienced volatility spikes sharply in 2008 and 2020.

We saw a persistence anomaly as the fit for FI collapses to near-unconditional levels in 2022, driven by sharp federal rate hikes.



Copula Fitting

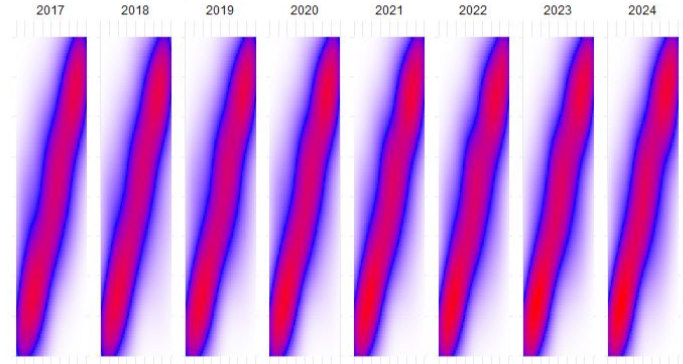
Equity and Private Equity maintain a consistent strong positive correlation ρ (.844, .859).

When equity experiences a massive sell-off, private equity follows.

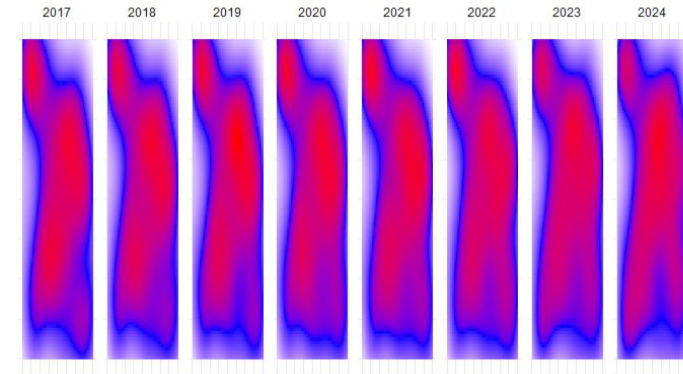
We see an initially negative correlation between Equity and Fixed Income deteriorate from $\rho \approx -0.3$ to $\rho < -0.1$.

Rate hiking cycle broke the historical inverse relationship, impacting flight-to-safety techniques.

EQ-PE Copula Density 2017-2024



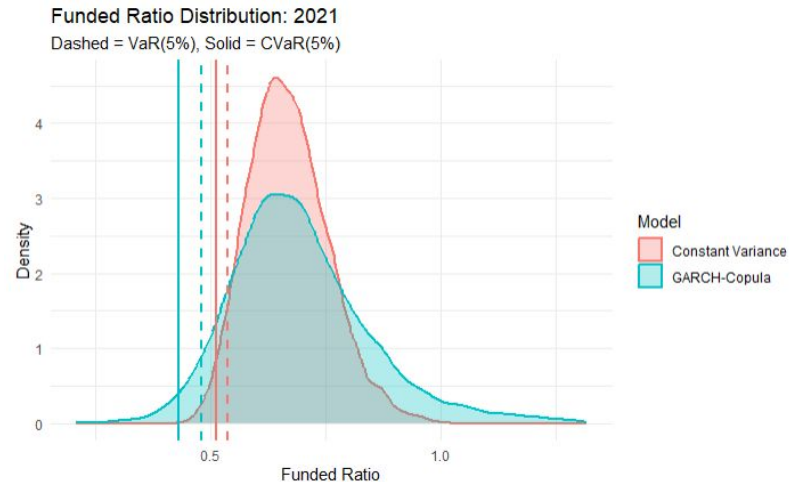
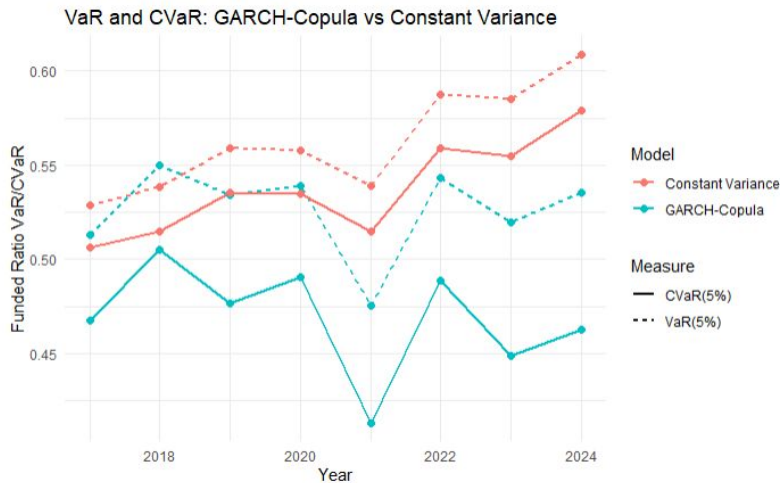
EQ-FI Copula Density 2017-2024



Clustered vs. Constant Volatility: A Monte Carlo Comparison

Clustered volatility gives a more conservative VaR & ES than a constant volatility assumption, averaging a mean VaR/cVaR gap of 4.8 and 7.4 points respectively.

COVID-19's effects drove the trough in 2021, while 2021's climb predicted relief that was bashed by 2022's crash.



PCA GARCH Alternative Methodology

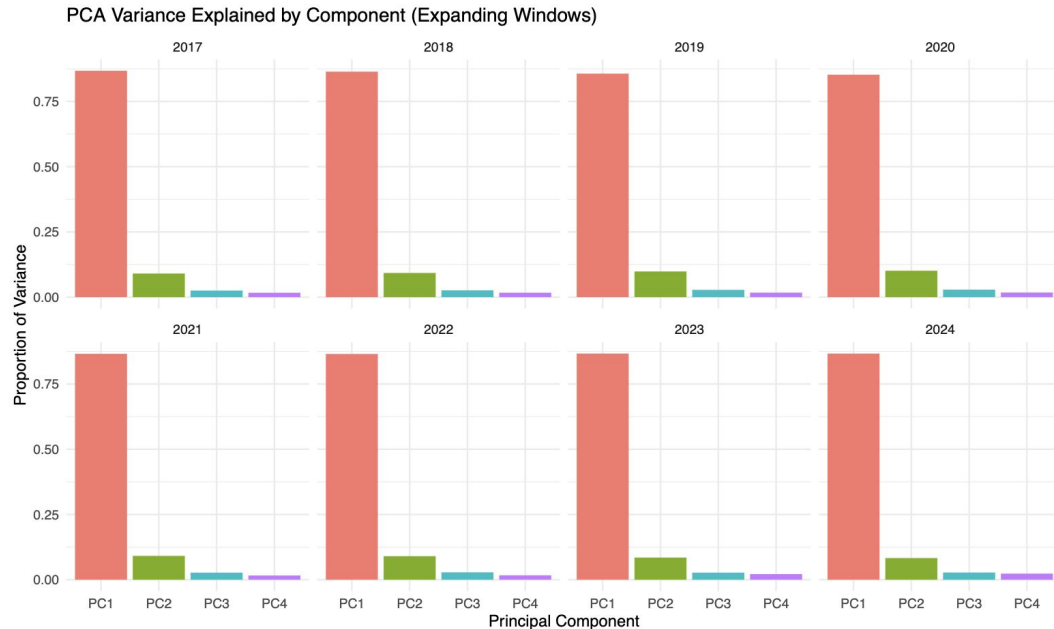
Pension portfolios hold 4 correlated asset classes.

Need a model that captures both correlation and time-varying volatility.

PCA rotates correlated returns into uncorrelated components with correlation stored in rotation matrix.

GARCH fitted to each component independently, captures volatility clustering.

Rotating back at simulation time automatically reproduces correlated asset returns.

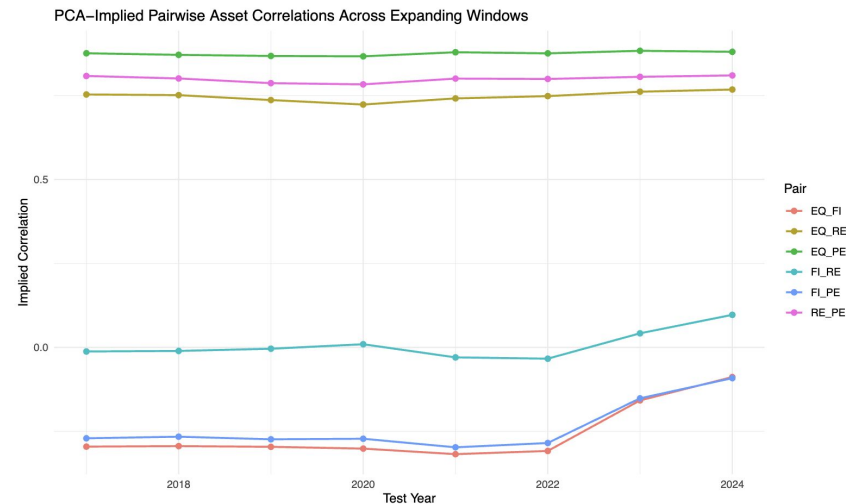
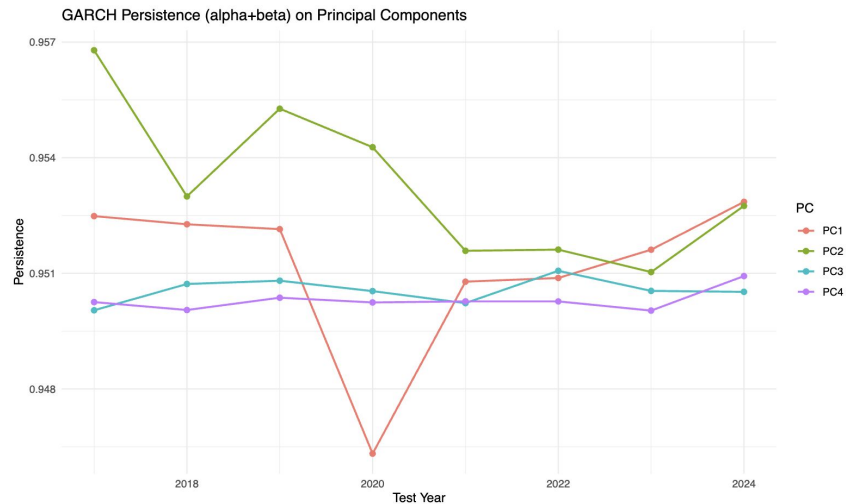


PCA GARCH Findings

PC1 is the broad market factor: Loads positively on all risky assets, a single factor drives the majority of portfolio variance, implying pension funds are not diversified.

High GARCH persistence: Volatility shocks are long-lived and the model conditions on current market state.

EQ-FI correlation drifts toward zero: The 2022 rate hiking cycle breaks the historical bond-equity hedge, where historical bonds would rally when equity falls.

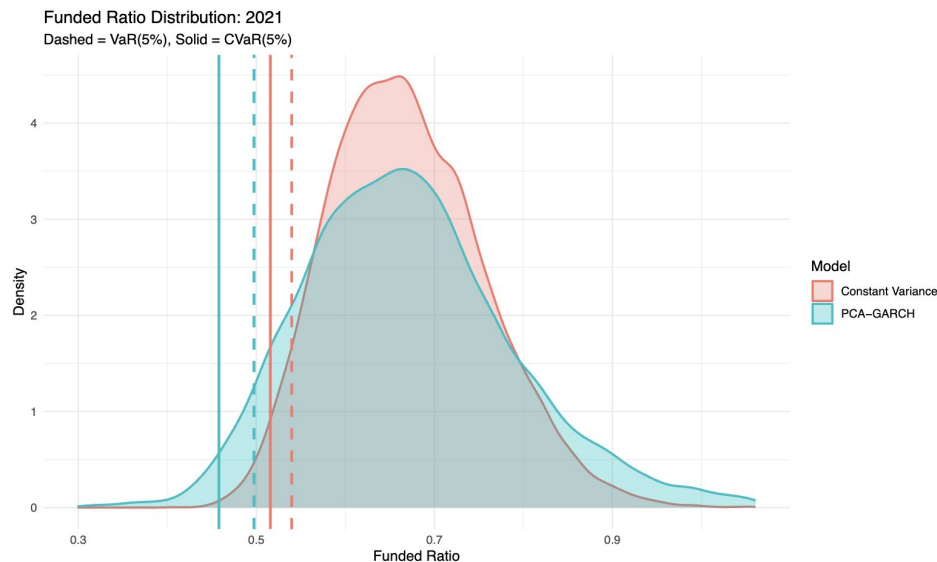
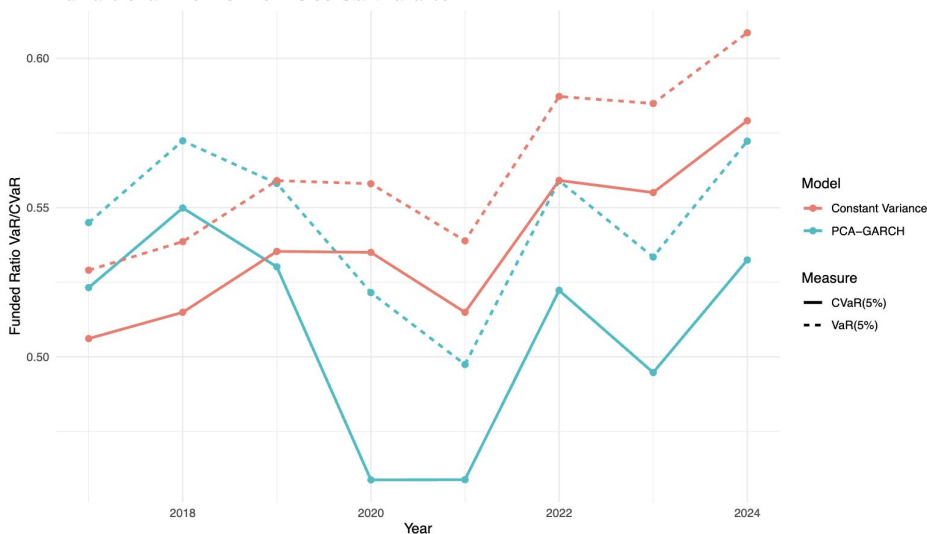


PCA GARCH Monte Carlo Results

Apply PCA GARCH model to compute VaR and CVaR in same way.

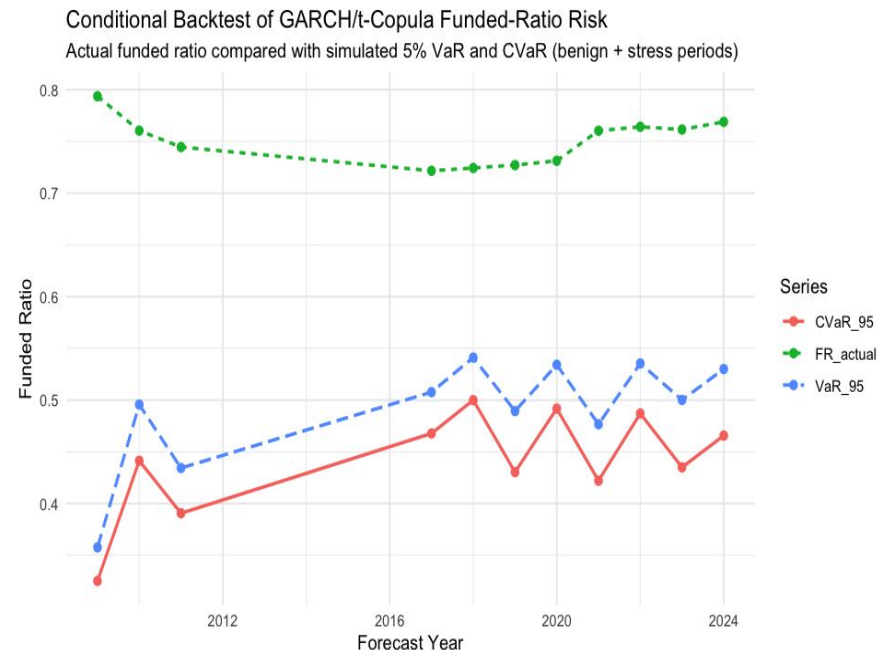
Model is able to capture downside risk better than the constant variance model.

VaR and CVaR: PCA-GARCH vs Constant Variance



Out-of-Sample Backtest of GARCH/t-Copula

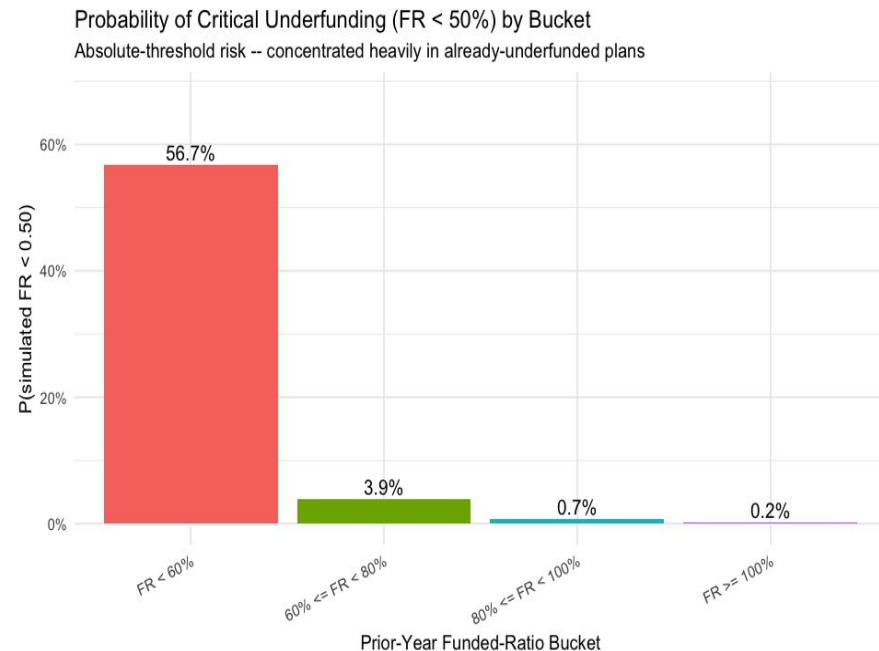
- GARCH-copula and PCA-GARCH both produce simulated risk distributions; we now compare them to realized funded ratios.
- Refit on data through year XX-1, simulate forward, compare with the realized FR in year XX.
- 11 forecast years tested, including the 2008-2009 crisis. Zero VaR breaches.
- In stress years the model widened its forecast tails; realized FRs landed near simulated medians, confirming correct conditioning on the post-crash regime.



Realized funded ratio (green) vs simulated 5% VaR (blue) and CVaR (red), 2009-2024.

Group-Level Funding Risk by Starting Bucket

- Bucketed plans by prior-year funded ratio; ran the same simulated scenarios per bucket.
- Level metrics scale near-mechanically: proportional drawdown ~ 0.72 across all four buckets.
- $P(\text{FR} < 50\%)$ is concentrated in the $\text{FR} < 60\%$ bucket (57%) and near zero for fully-funded plans (0.2%) — a 280x range.
- Well-funded plans face the largest absolute drawdowns (-30 pp at 5% tail) but stay far above critical levels.



*$P(\text{simulated FR} < 0.50)$ by prior-year funding bucket.
Catastrophic-threshold risk concentrates in already-underfunded plans.*

Limitations

Liabilities are anchored to reported values and adjusted using a fixed duration. True liability is stochastic, however 24 data points is too few to fit a model. A Bayesian informative prior was tested with no significant improvement.

The PCA analysis assumed a Gaussian structure, likely explaining the optimistic estimate compared to the t-copula.

For numerical stability we constrained extreme portfolio gains, since a small number of paths were positively explosive, which were realistically not feasible.

PPD returns are actuarially smoothed, likely leading to overstated volatility on the asset side, with understated volatility on the liability side.

Summary

We see an increasing flight from equity following market crashes.

Historical correlation between asset types change as unprecedented economic scenarios emerge, notably equity and fixed income.

Volatility shock carries over time, and modelling this accounts for greater risk, providing conservative predictions.

Although the funded ratio stays above the VaR for the simulated years, a 1 in 20 year result will prove significantly problematic for pension plan funding.

In general, pension plans assume historical results, resulting in under-diversification in new regimes and unprecedented shocks. This enforces the public's growing concern for state and local retirement security.

For a fund of NYPD's size a 7.4 point cVaR gap can result in an understated loss of \$4.4 billion.